

Scattering. Should be around a similar point in the book.

Scattering is very important for understanding a vast variety of subjects, especially condensed matter. You can't escape it. Today's class felt like a coherent lesson with direction.

Files:

0) solid angle.pdf (continued from yesterday)

1) scattering cross section.pdf

2) cross sections demo MC.nb

3) hard sphere scattering.pdf

Key ideas:

- Impact Parameter
The closest perpendicular/asymptotic approach from a long distance. Corresponds to the angular momentum that is conserved.
- Scattering angle
Deviation between incoming/outgoing vector
- Cross Section
differential area & cross section discussed today
- Solid angle Ω
(discussed on 9/22 lecture)
 $\Omega \sim \frac{\text{Area}}{R^2}$ but not exactly

$$d\Omega = \frac{\hat{r} \cdot d\vec{A}}{r^2}$$
$$\Omega = \int_{\text{surface}} d\Omega$$

Max (in 3D) is 4π , when a source is completely surrounded.

See 0) solid angle.pdf for source in sphere/outside sphere.

source location r' (1)

sphere surface $d\vec{A} = \hat{r} \cdot 2\pi R \sin(\theta) R d\theta$ (2)

integrate! (3)

In the pdf Bunker pulls out a $\frac{d}{dr}$ to make computing the integral easier, essentially pattern recognition.

Example:

$$\begin{aligned}
 \int_{-\infty}^{\infty} x^2 e^{-ax^2} dx &= -a \frac{d}{da} \int_{-\infty}^{\infty} e^{-ax^2} dx = -a \frac{d}{da} I \\
 I^2 &= \int_0^{\infty} e^{-ax^2} dx \int_0^{\infty} e^{-ay^2} dy \\
 &= \int_0^{\infty} \int_0^{\infty} e^{-a(x^2+y^2)} dx dy \\
 &= \sqrt{\frac{\pi}{a}}
 \end{aligned}$$

Then go back to I , or something idk. There's a few other math tricks there too.

Ω is zero for $a > R$ since the surface facing towards and away cancel, and this integration matches the shell theorem / gauss's theorem

Point near surface: $\Omega = 2\pi$

Far from surface: Still 2π if the surface is infinite. See document for calculation of non-infinite surface (circle, square).

Note on notes: For rectangle surface, the geometric mean is $\sqrt{(\text{nearest distance})(\text{furthest distance})} = \sqrt{dd'}$, so squaring that value results in the value for rectangle=square.

Cross Section

Encountered across fields, but boils down to interaction probability. Spin-spin interactions, baseballs through dust clouds, it's kind of the same. Units of area.

Discussion in class mostly lines up with Taylor's description.

Discussion from 1) scattering cross section.pdf:

$$N_{\text{scatter}} = N_{\text{incident}} \cdot \underbrace{n_{\text{target}}}_{\text{target area density}} \cdot \underbrace{\sigma}_{\text{cross section}}$$

σ can depend on parameters, including scattering angle.

In the diagram on page two, note $l = |\vec{r} \times \vec{p}| = pb$, momentum p times impact parameter b is conserved. Angular momentum pops out of the spherical coordinates lagrangian as

$$l = \frac{\partial L}{\partial \dot{\phi}} = \mu r^2 \dot{\phi}$$

which becomes relevant on page 3.

Differential cross section $\frac{d\sigma}{d\Omega}$ is used to consider various parameters of σ .

Differential Cross Section

See 2) cross sections dem MC.nb

What is the mathematica notebook doing? First, generates a bunch of points (blue vs red) and checks their overlap.

From there, the notebook has:

- probability of one red particle hitting one blue
- exact probability through analytic integral

The result matches the formula above ($N_{\text{scatter}} = \dots$), and yields the cross sectional area $\sigma = \frac{\pi R^2}{4}$. Unfortunately, Bunker does not obey Donald Knuth and then goes on to find the error due to finite sampling. Also ρ is used instead of σ . The professor suggests it could be a good way to handle edge effects with detectors.

Last 7 minutes of class: hard sphere potential

See 3) hard sphere scattering.pdf

Donald Knuth (TeX) says premature optimization is the root of all evil (and may introduce bugs -Bunker)